Categorical Preference Analysis

Chi-square analysis in R

Executive Summary

This analysis examined categorical flavour preferences for 100 participants. Overall, 54 participants preferred chocolate and 46 preferred strawberry. That split did not differ significantly from an equal 50/50 distribution, chi-square(1) = 0.64, p = .424.

Preference was not distributed equally across gender. Men selected chocolate 20 times and strawberry 28 times; women selected chocolate 34 times and strawberry 18 times. A chi-square test of independence found a significant association, chi-square(1) = 4.74, p = .0295. Adjusted residuals of approximately plus or minus 2.38 identified the specific cells driving the association.

100	54%	46%	.0295
Participants	Chocolate preference	Strawberry preference	Association p-value

Research Questions

1. Does the overall flavour split differ from a specified expected distribution?
2. Is flavour preference associated with participant gender?
3. Which cells in the contingency table contribute most strongly to any association?

Observed Data

Gender	Chocolate	Strawberry	Row total
Men	20	28	48
Women	34	18	52
Column total	54	46	100

The analysis used frequency tables, goodness-of-fit tests, a test of independence and residual analysis. These methods are appropriate because the variables are categorical counts rather than continuous measurements.

Analytical Workflow

- Created one-dimensional and two-dimensional frequency tables in R.
- Calculated proportions to distinguish raw counts from relative frequency.
- Compared observed overall preferences with equal and custom expected distributions.
- Tested the gender-preference contingency table with Pearson's chi-square test using Yates' continuity correction.
- Inspected expected frequencies and adjusted residuals to locate the cells driving the significant result.
- Used bar charts and the chi-square distribution to support interpretation.

Gender and Flavour Preference

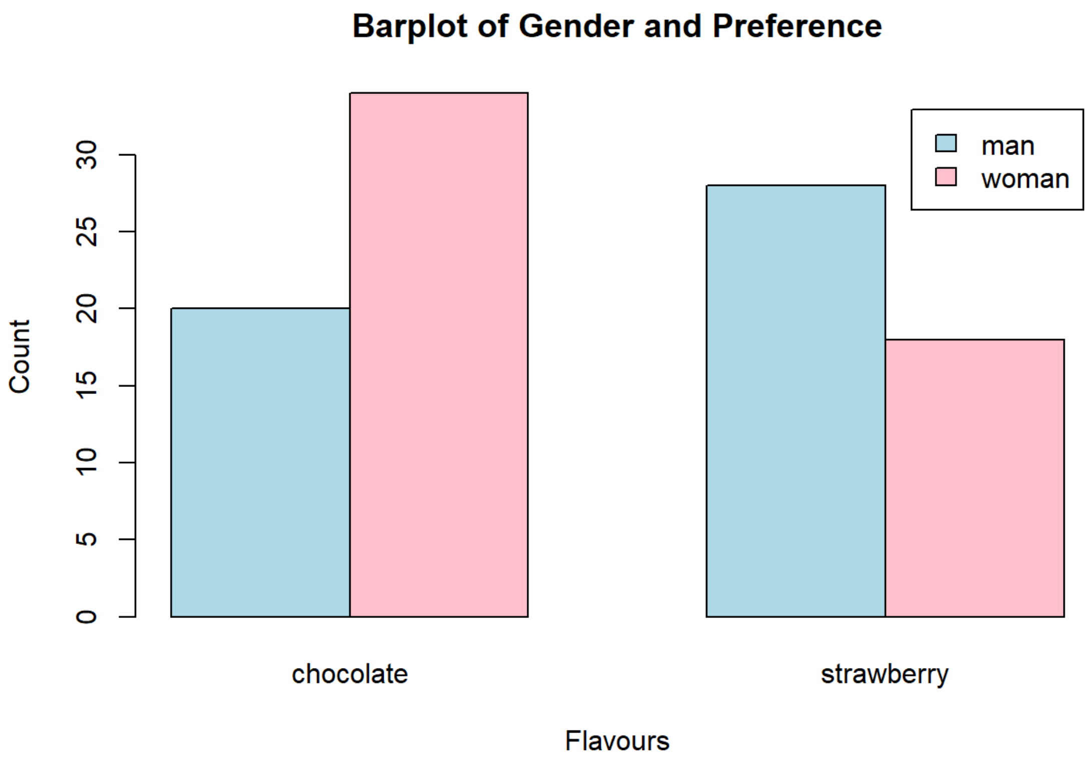

Figure 1. Counts by gender and flavour preference.

> **Visual interpretation**
> The direction of the association is visible before the formal test. Women selected chocolate almost twice as often as strawberry (34 versus 18), whereas men selected strawberry more often than chocolate (28 versus 20). The crossover pattern suggests that preference proportions differ by gender rather than reflecting one flavour being uniformly more popular for everyone.

Overall Goodness-of-Fit Test

Comparison	Observed	Expected	Test result	Conclusion
Equal split	54 / 46	50 / 50	chi-square(1) = 0.64, p = .4237	No significant deviation
Custom split	54 / 46	40 / 60	chi-square(1) = 8.17, p = .0043	Significant deviation

The same observed data can produce a different conclusion when the expected distribution changes. Against a 50/50 benchmark, a four-person difference is compatible with random variation. Against a 40/60 benchmark, the observed 54/46 split is far enough away to reject the expected pattern.

Observed Versus Equal Expected Counts

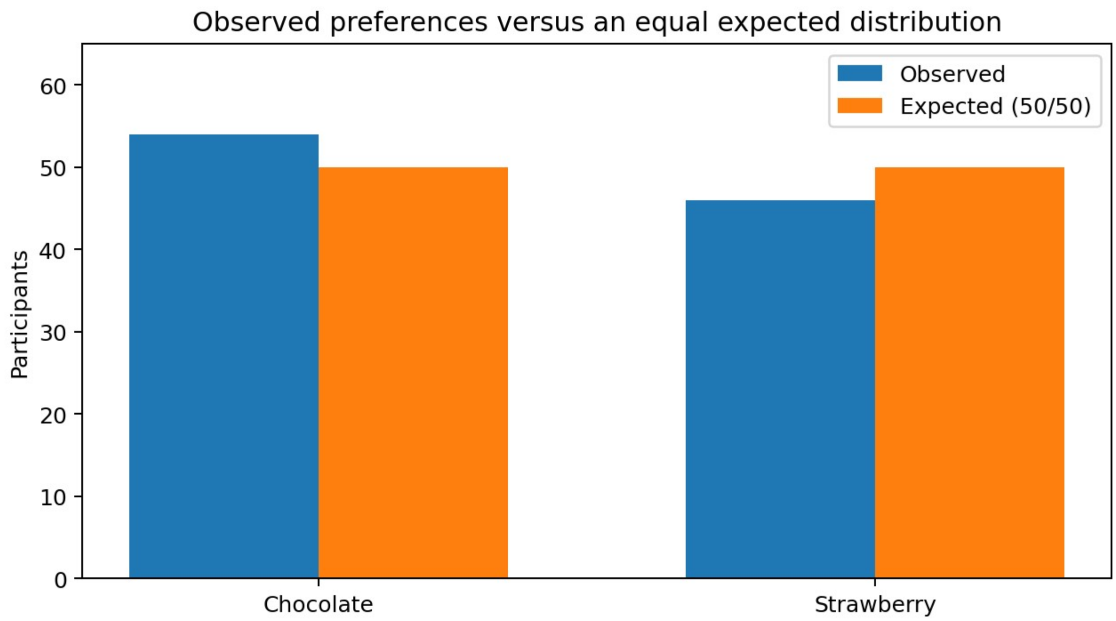

Figure 2. Overall observed preferences compared with an equal expected distribution.

Visual interpretation

The bars differ by only four participants in each direction from the equal expectation. Relative to the total sample of 100, this gap is modest, which is consistent with the non-significant p-value of .4237.

Test of Association

Statistic	Value	Interpretation
Pearson chi-square with Yates correction	4.7381	Observed table differs from independence
Degrees of freedom	1	2 x 2 contingency table
p-value	.0295	Statistically significant at alpha = .05
Phi / Cramer's V	0.218	Small-to-moderate association

The significant test indicates that knowing gender provides some information about flavour preference in this sample. The effect size is approximately 0.218, so the association is meaningful enough to detect but not strong enough to treat gender as a deterministic predictor.

Adjusted Residual Analysis

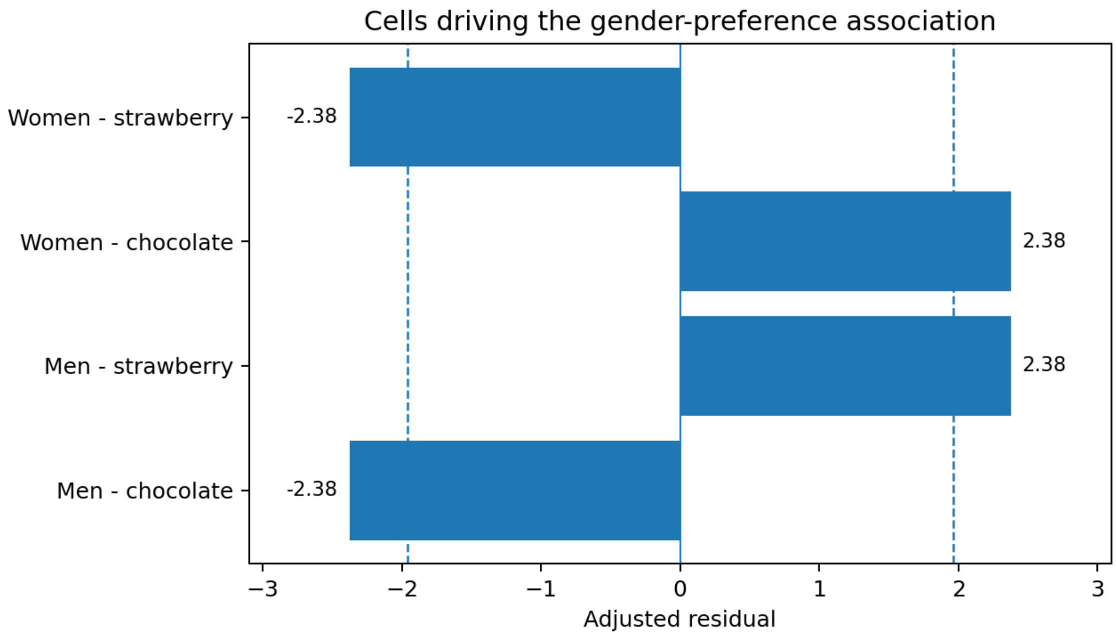

Figure 3. Adjusted residuals for each cell; dashed reference lines mark plus or minus 1.96.

Visual interpretation

All four residual magnitudes exceed 1.96. Positive residuals indicate more observations than expected, while negative residuals indicate fewer. Men selected strawberry more often than expected (+2.38) and chocolate less often than expected (-2.38). Women showed the reverse pattern: more chocolate (+2.38) and less strawberry (-2.38). This explains where the overall association originates.

Conclusion

There was no evidence that chocolate and strawberry preferences departed from a simple equal split across the entire sample. There was, however, evidence that the pattern differed by gender. The report therefore separates two distinct questions: overall popularity and association between variables.

Limitations and Next Steps

- The data are educational and categorical; they do not establish why participants preferred a flavour.
- The analysis includes only two gender categories and two flavour categories, which limits representation and scope.
- A larger study could collect additional preference categories and report confidence intervals for proportions.
- Predictive use would require out-of-sample validation rather than relying on one contingency table.